Data Analyst Capstone Project: Developer Technology Trends

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IBM Data Analyst Professional Certificate



OUTLINE



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EXECUTIVE SUMMARY



Key findings from the 2024 Stack Overflow Developer Survey (65,437 developers) and job-market data:

- **Languages:** JavaScript is the most used today (37.5K); Python is the most wanted for next year (25.0K). Rust and Go enter the wanted top 10.
- **Databases:** PostgreSQL is #1 now (25.5K) and next year (24.0K). SQLite rises; MS SQL Server and Oracle lose interest.
- **Job market:** Washington DC has the most job postings (5,316). Swift pays the highest average salary ($130.8K), PHP the lowest ($84.7K).
- **Pay:** rises with experience ($27K at 0–2 yrs → $106K at 31+ yrs) and depends most on country (US ≈ $130K vs India ≈ $17K median). Satisfaction barely tracks pay (r = 0.08).
- **Who answered:** 68.6% are aged 25–44 and 72% hold a Bachelor's or Master's degree; the US is the largest group.

INTRODUCTION



Purpose

- Identify which programming languages, databases and platforms developers use today and want to use next year — and what drives pay and job satisfaction.

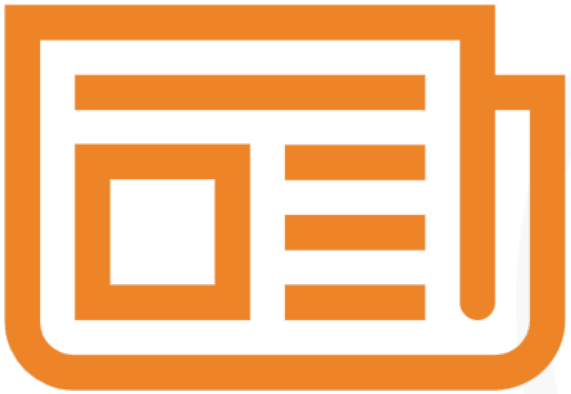
Target audience

- Hiring managers and tech recruiters planning which skills to hire for
- Technology leaders choosing a tech stack and training budget
- Developers and learners deciding which skills to build next

Value of the analysis

- Data-backed view of where skill demand is heading, so hiring, training and career decisions follow the market instead of guesswork.

METHODOLOGY



Data sources

- Stack Overflow Developer Survey 2024 (CSV, 65,457 rows × 114 columns)
- Job postings API → job-postings.xlsx (postings by location and technology)
- Web-scraped salary table → popular-languages.csv (average salary by language)

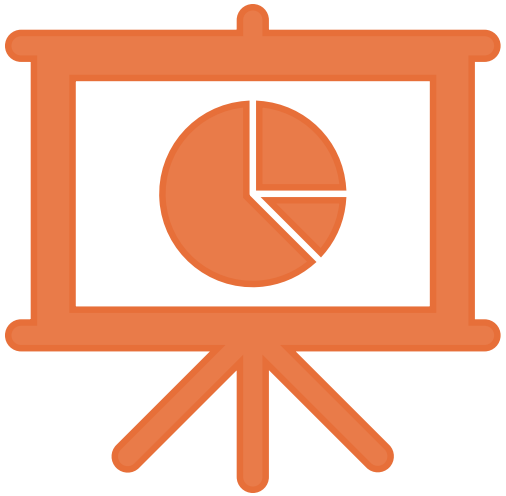
Data collection

- Python: requests + JSON for the API, BeautifulSoup for scraping, pandas to load the survey

Data wrangling & analysis

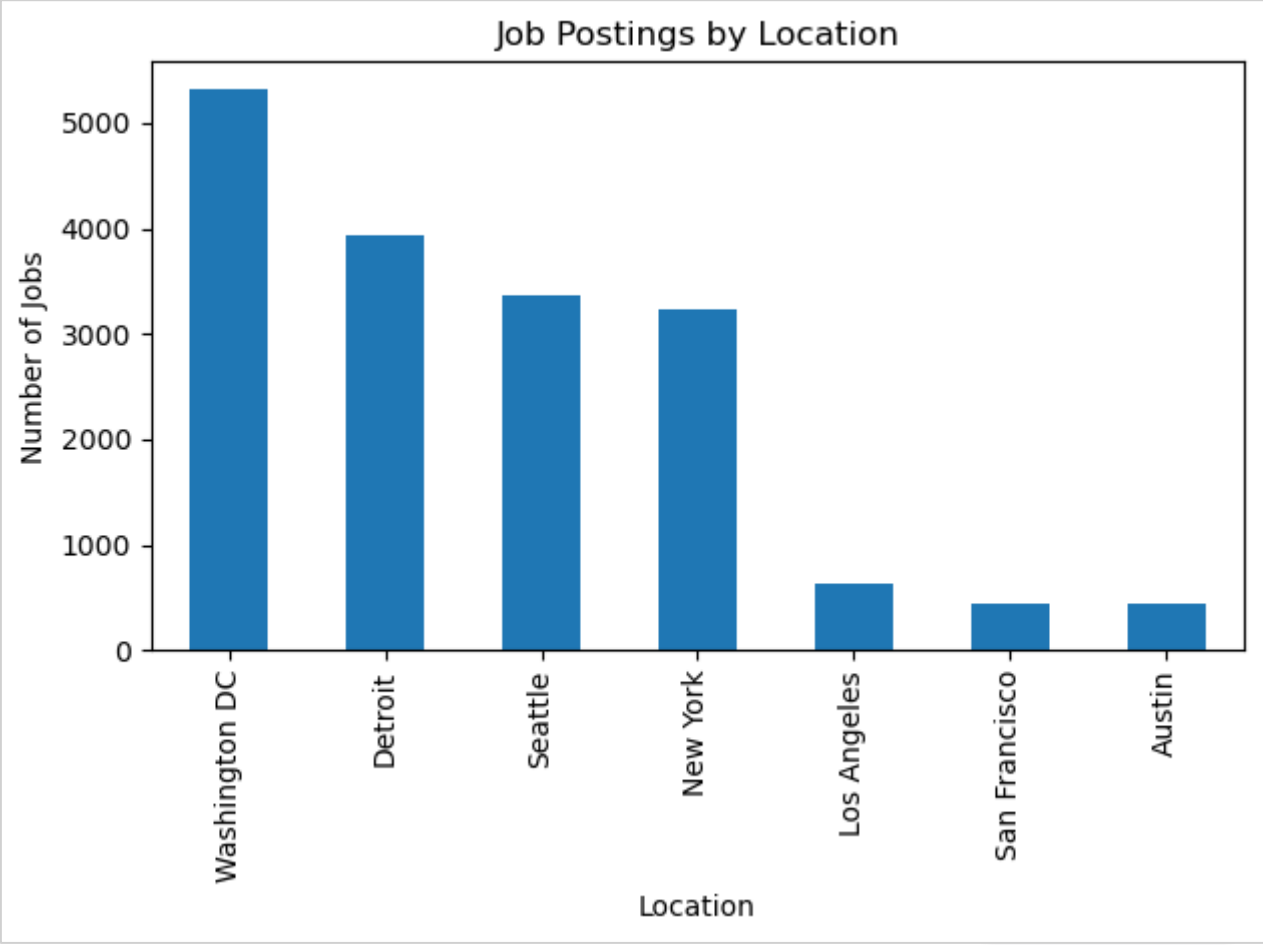
- Removed 20 duplicate rows (→ 65,437); handled missing values (mode imputation for categories; salary analysed on real values only)
- Normalized salary (Min-Max, Z-score, log); removed 978 IQR outliers above $220,861
- Split multi-select answers (";") with explode; grouped ages and experience into ranges
- Charts with matplotlib/seaborn; dashboard in Google Looker Studio (lab sample of 18,845 rows)

Results



Job market, programming language and database trends

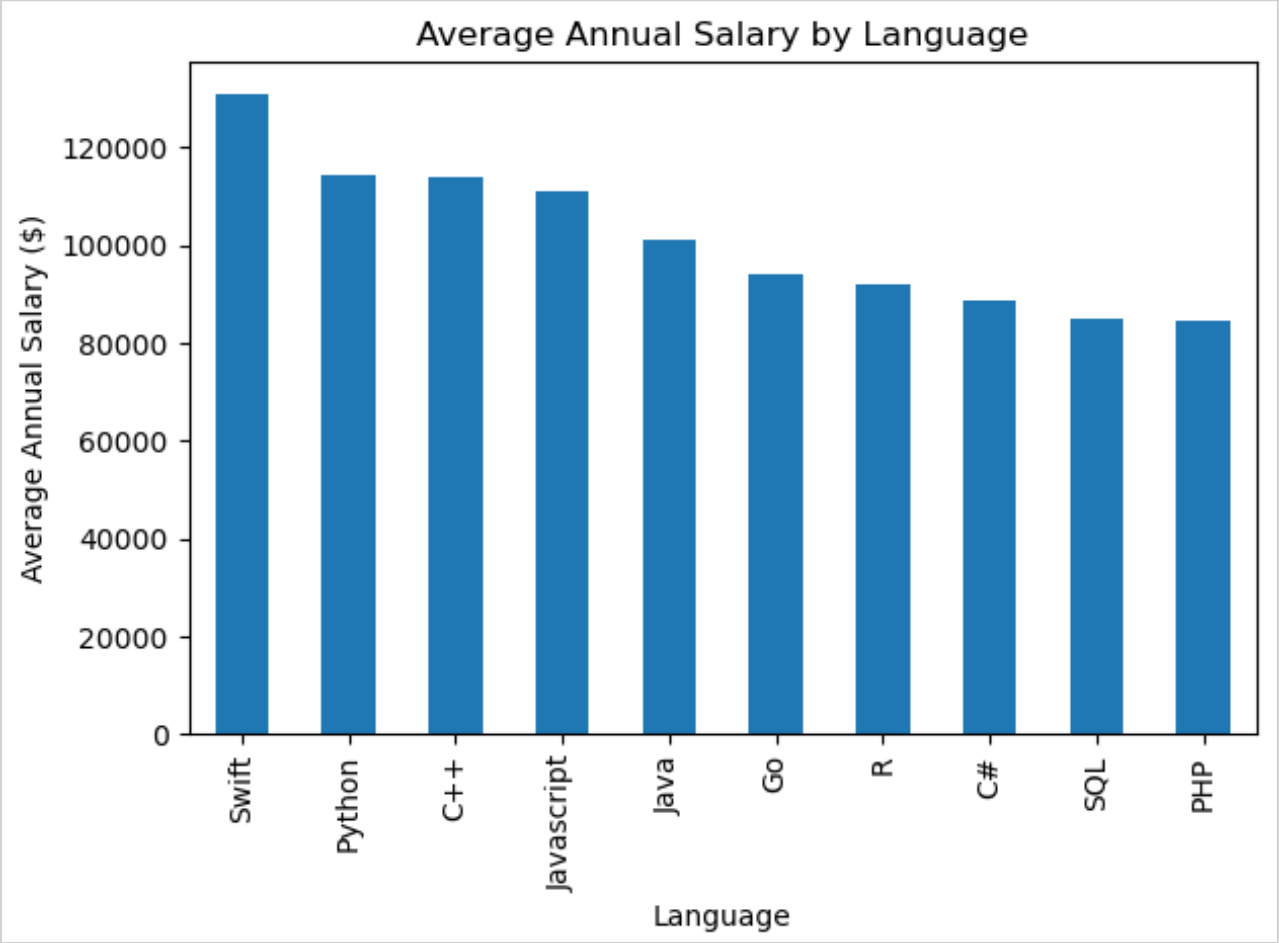
JOB POSTINGS



- Washington DC leads with 5,316 job postings
- Detroit (3,945), Seattle (3,375) and New York follow
- Los Angeles, San Francisco and Austin have far fewer postings

Source: job postings API (Module 1), sorted by number of postings, descending

POPULAR LANGUAGES



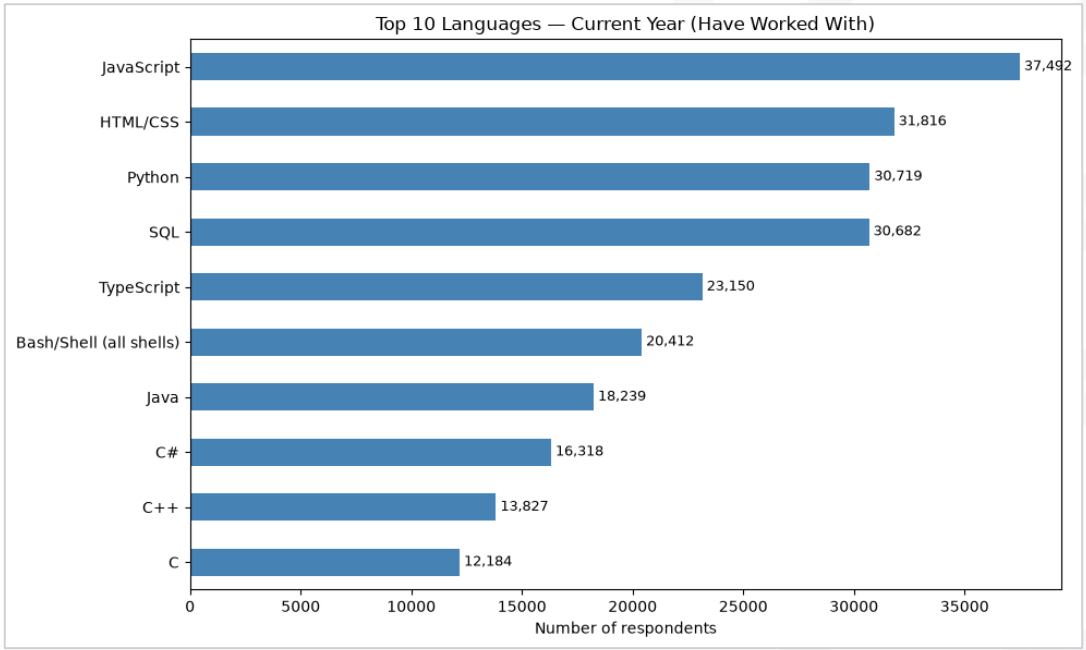
- Swift has the highest average salary: $130,801
- Python, C++ and JavaScript all average above $110K
- PHP has the lowest: $84,727

Source: web-scraped salary table (Module 1), sorted by average salary, descending

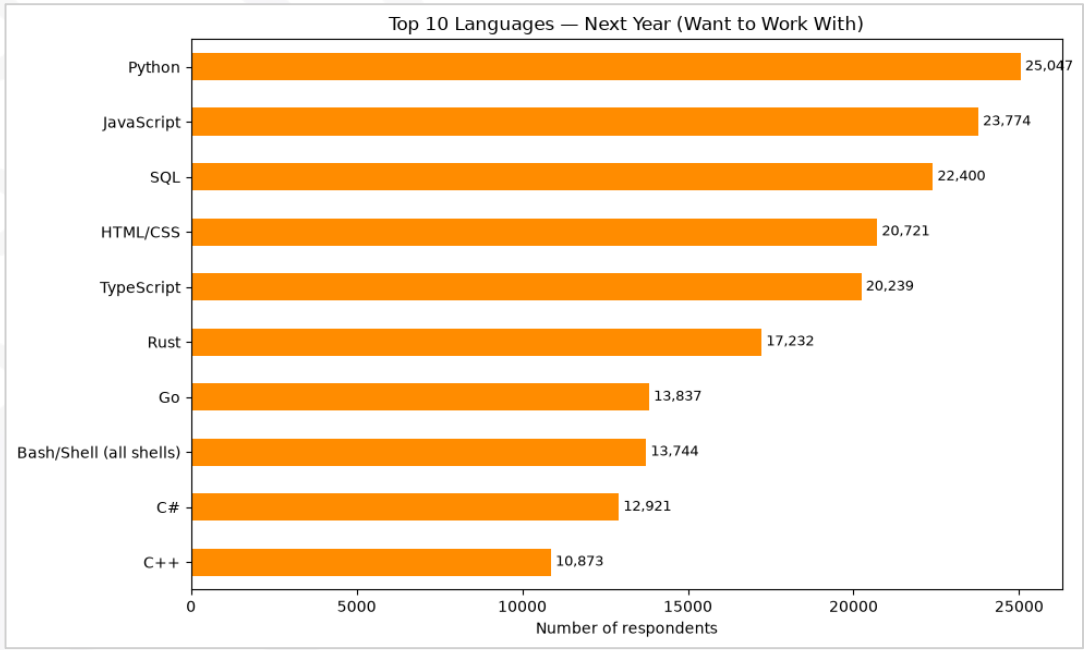
PROGRAMMING LANGUAGE TRENDS

JavaScript leads today, but Python is #1 for next year. Rust and Go join the top 10, while Java and C drop out.

Current Year



Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- JavaScript is the most used language (37,492), but Python is the most wanted for next year (25,047).
- Rust (17,232) and Go (13,837) are not in today's top 10 but enter the wanted top 10.
- Java and C drop out of the wanted top 10; SQL stays in the top 4 in both years.

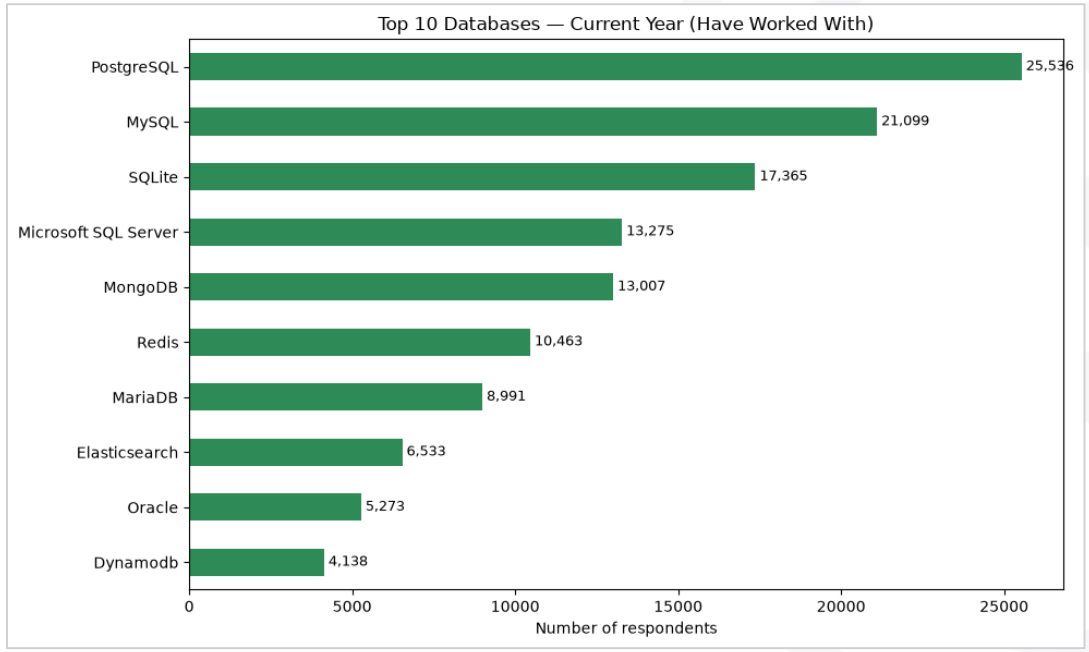
Implications

- Python skills will be in the highest demand as data, AI and automation work keeps growing.
- Developers are moving to fast, memory-safe modern languages — early adopters gain an edge.
- SQL is a stable core skill for every role; legacy Java/C code still needs support but attracts less new interest.

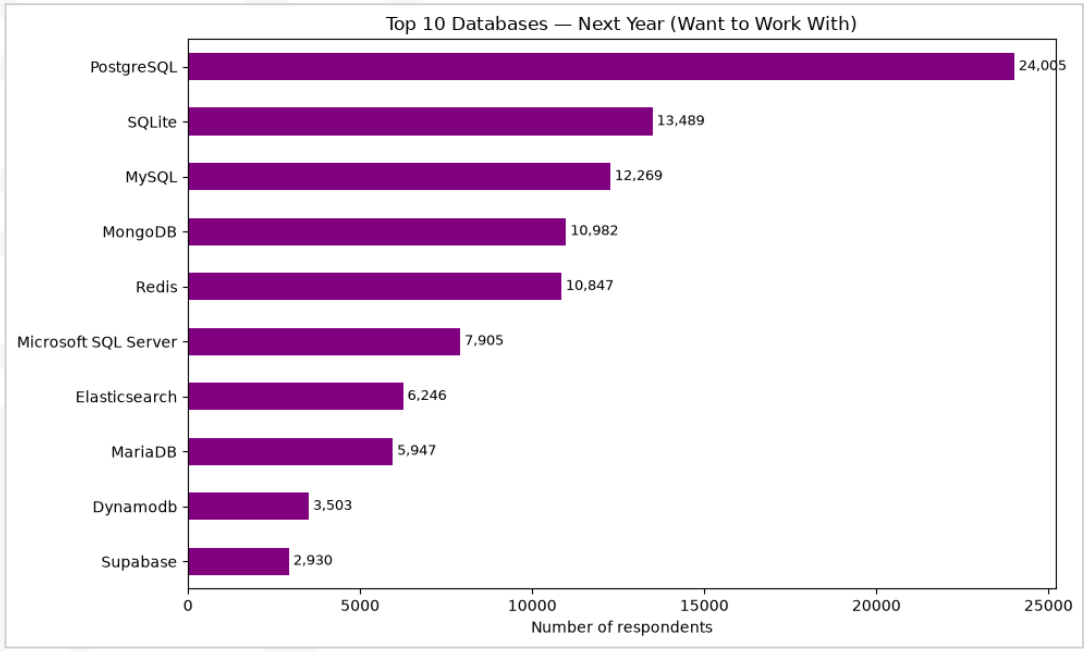
DATABASE TRENDS

PostgreSQL is #1 in both years. SQLite rises to #2, while MySQL, MS SQL Server and Oracle lose interest; Supabase enters.

Current Year



Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL is #1 now (25,536) and next year (24,005) — almost everyone who uses it wants to keep it.
- SQLite rises to #2 and MySQL falls to #3; MS SQL Server interest drops from 13,275 to 7,905.
- Oracle drops out and Supabase (cloud-hosted PostgreSQL) enters the wanted top 10.

Implications

- PostgreSQL is the safest database to learn and to standardize on.
- Open-source and lightweight databases are gaining over commercial, licensed ones.
- Cloud-hosted, developer-friendly database services are a growing skill area.

DASHBOARD

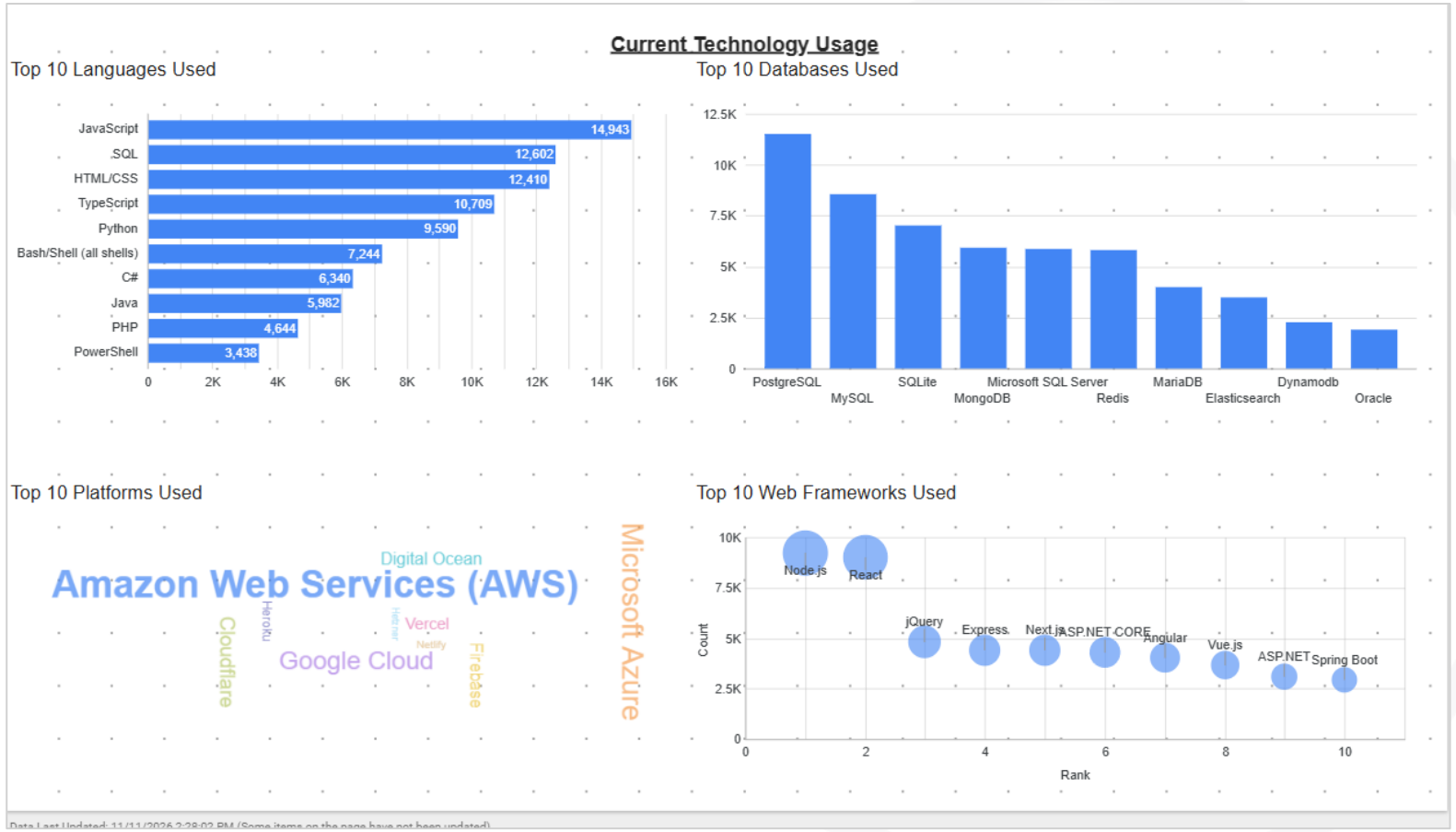


Google Looker Studio dashboard with 3 tabs:

- Current Technology Usage
- Future Technology Trends
- Demographics

Live dashboard: https://datastudio.google.com/reporting/08c01067-2253-41c3-9d22-89f28c48fdf4

DASHBOARD TAB 1: Current Technology Usage

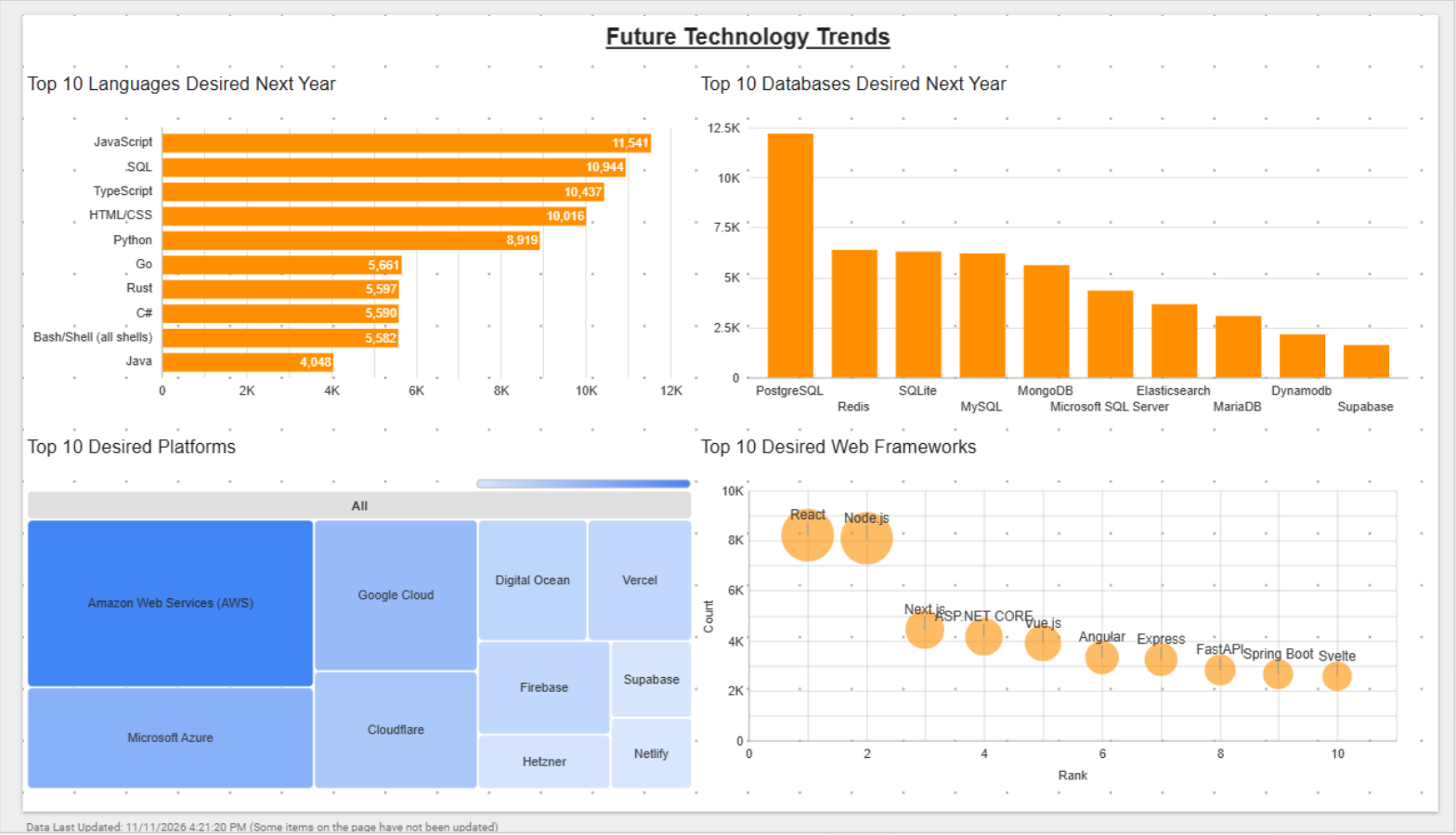


Dashboard Insight

- JavaScript is the most used language (14,943)
- PostgreSQL is the top database (11,514)
- AWS is the most used cloud platform (10,871)
- Node.js (9,230) and React (8,999) lead web frameworks

Dashboard data: lab sample of the survey (18,845 respondents), top 10 per chart

DASHBOARD TAB 2: Future Technology Trends

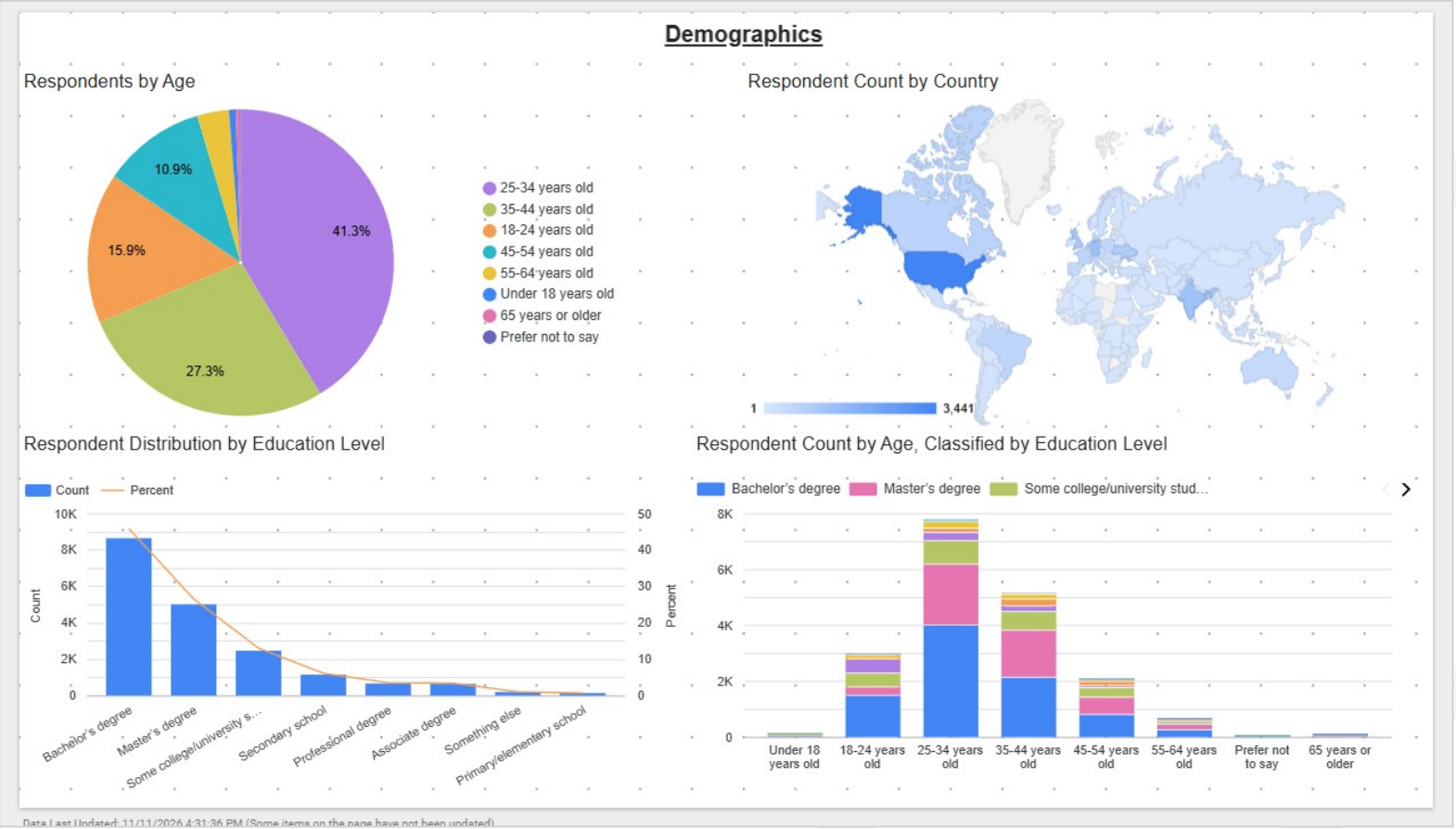


Dashboard data: lab sample of the survey (18,845 respondents), top 10 per chart

Dashboard Insight

- JavaScript (11,541), SQL (10,944) and TypeScript (10,437) are the most desired languages
- Go and Rust enter the top 10; Java falls to #10
- More developers want PostgreSQL (12,193) than use it today (11,514)
- React (8,223) overtakes Node.js (8,096)

DASHBOARD TAB 3: Demographics



Dashboard data: lab sample of the survey (18,845 respondents), top 10 per chart

Dashboard Insight

- 25–34 is the largest age group (41.3%); 68.6% are aged 25–44
- The United States has the most respondents (3,441), across 161 countries
- Bachelor's (45.8%) and Master's (26.5%) cover 72% of respondents
- Bachelor's is the largest group in almost every age band

DISCUSSION



Overall insights from the dashboard

- **Stable core:** JavaScript, SQL and PostgreSQL rank at the top in both the current and the desired views.
- **Shift to modern tools:** TypeScript, Go, Rust, React and PostgreSQL gain in the desired view; Java falls.
- **Cloud:** AWS is the most used platform, so cloud skills sit next to languages and databases.
- **Who is speaking:** mostly 25–44, degree holders, US-heavy — the trends reflect working professionals.
- **Limitations:** self-reported survey; multi-select answers show popularity, not market share.

OVERALL FINDINGS & IMPLICATIONS

Findings

- The core stack is stable: JavaScript, SQL, Python and PostgreSQL lead both now and next year.
- Demand shifts toward modern, open-source tools (Rust, Go, TypeScript, SQLite, Supabase) and away from legacy/commercial ones (Java, C, Oracle, MS SQL Server).
- Pay rises with experience ($27K → $106K median) and depends most on country; satisfaction barely tracks pay (r = 0.08).

Implications

- Employers and learners should prioritize Python, SQL and PostgreSQL — safe, high-demand skills.
- Tech leaders should plan training and hiring for Rust, Go, TypeScript and cloud databases, and budget for moving off legacy commercial systems.
- Retaining developers takes more than salary — the work itself, growth and environment matter for satisfaction.

CONCLUSION



- JavaScript is today's most used language; Python is the top choice for the future.
- PostgreSQL is the leading database now and next year, and open-source databases are gaining.
- Job postings concentrate in a few cities (Washington DC, Detroit, Seattle); Swift, Python and C++ pay the highest average salaries.
- Experience and location drive pay the most — building in-demand skills (Python, SQL, PostgreSQL, cloud) is the best career investment.

APPENDIX

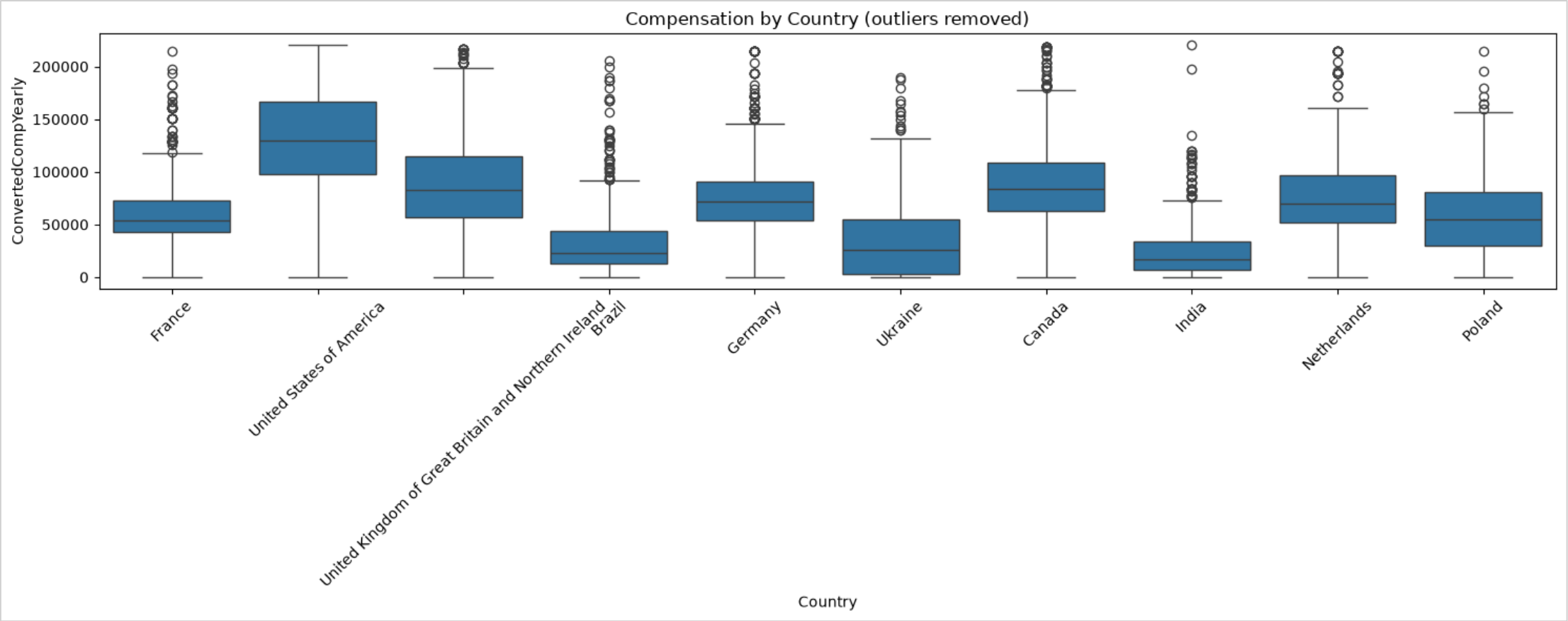


Additional charts created during the analysis phase:

- A1 — Compensation by country (outliers removed)
- A2 — Median compensation by years of experience
- A3 — Distribution of job satisfaction (0–10)

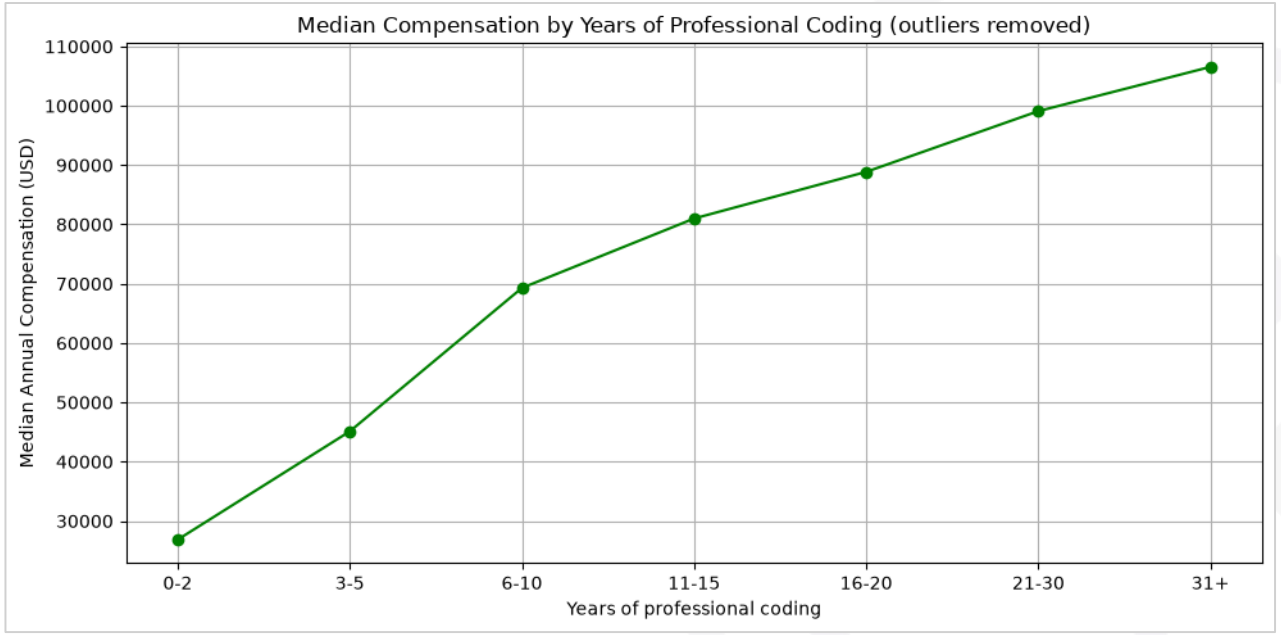
All charts built in Python (pandas, matplotlib, seaborn) from the 2024 Stack Overflow survey.

APPENDIX — A1

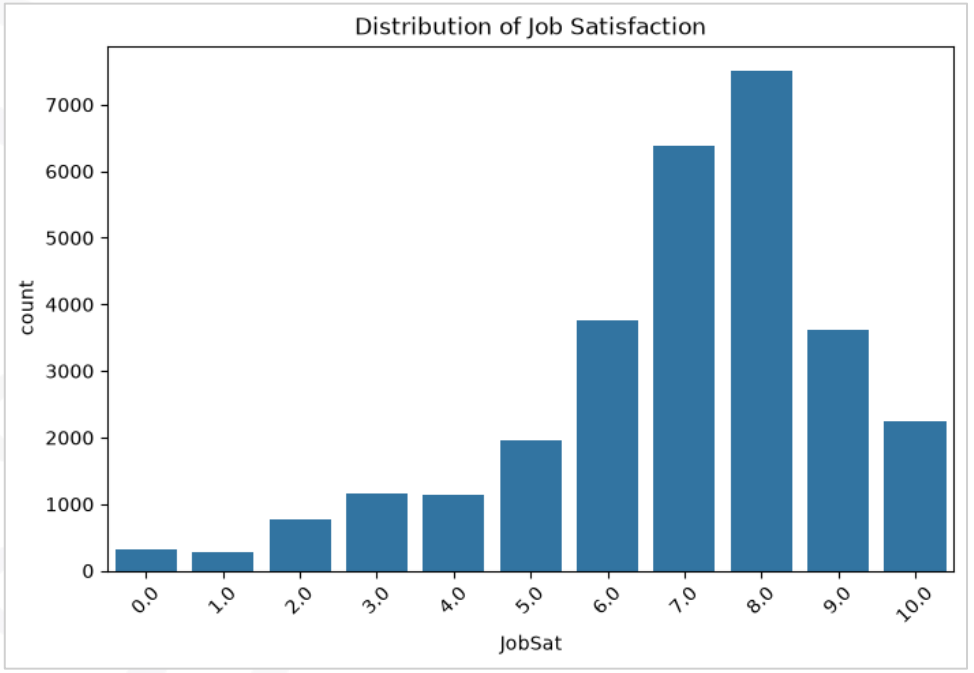


A1 — Compensation by country (outliers removed): US median ≈ $130K is the highest; India ≈ $17K the lowest.

APPENDIX — A2, A3



A2 — Median compensation rises with experience: ≈ $27K (0–2 yrs) → ≈ $106K (31+ yrs).



A3 — Most developers are satisfied: median 7, most common score 8.